

Rehabilitating the benefits of gene tree correction in the presence of ILS

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“Correcting” Gene Trees to be More Like Species Trees Frequently Increases Topological Error

Zhi Yan  , Huw A Ogilvie  , Luay Nakhleh 

Genome Biology and Evolution, Volume 15, Issue 6, June 2023, evad094,
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"we have demonstrated here that **species tree attraction methods should be used with extreme caution** when ILS causes the true gene tree histories to be highly discordant from the history of corresponding species"

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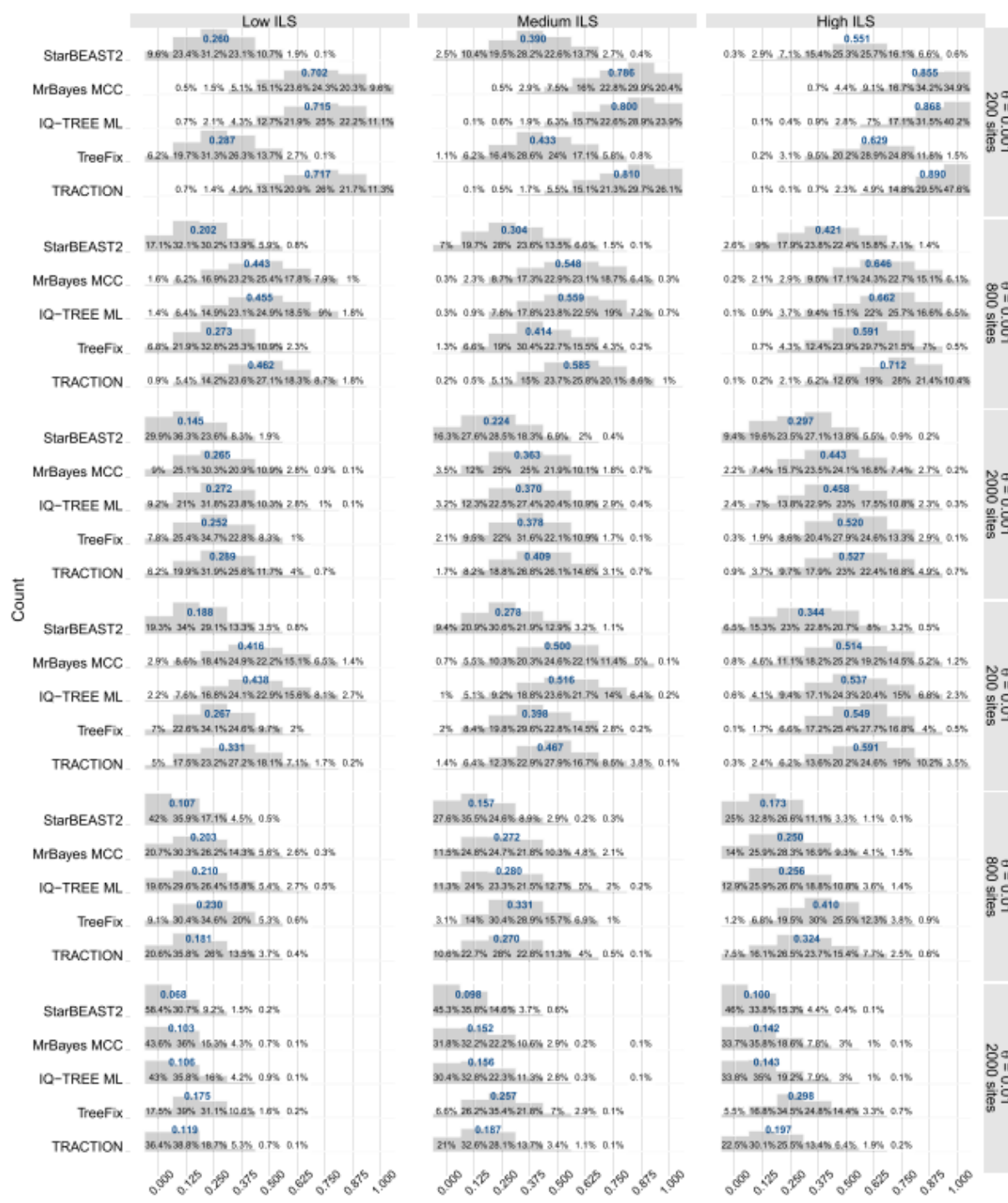
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Argument: ILS makes the gene tree differ from the species tree, so don't change it.



Simulations

- Multispecies coalescent
- One gene per species
- ILS only possible disagreement with species tree

TreeFix

- Starts with given gene tree
- Explores space with moves
- Finds gene tree with higher likelihood and similar dup/loss cost

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- Finds refinement at min RF distance from species tree

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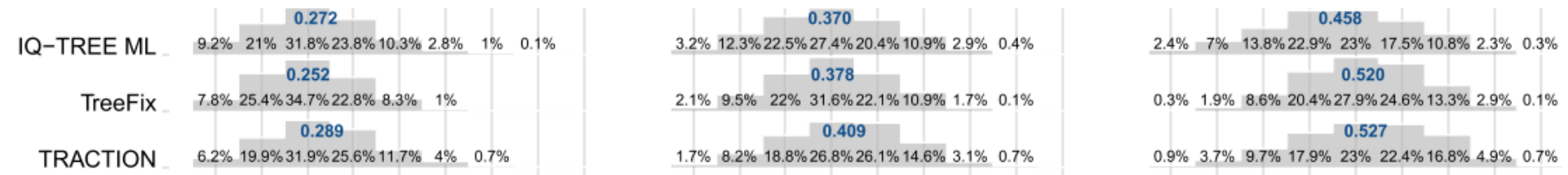
TRACTION

- Collapse low-support branches (bootstrap <70)
- Finds refinement at min RF distance from species tree
- Does not consider dup/loss events

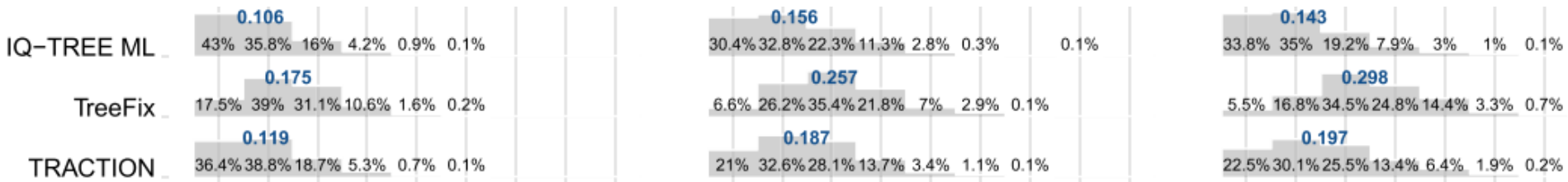
Average RF error

IQ-TREE ML: before correction

Low mutation rate



High mutation rate



good --> bad

ecceTERA

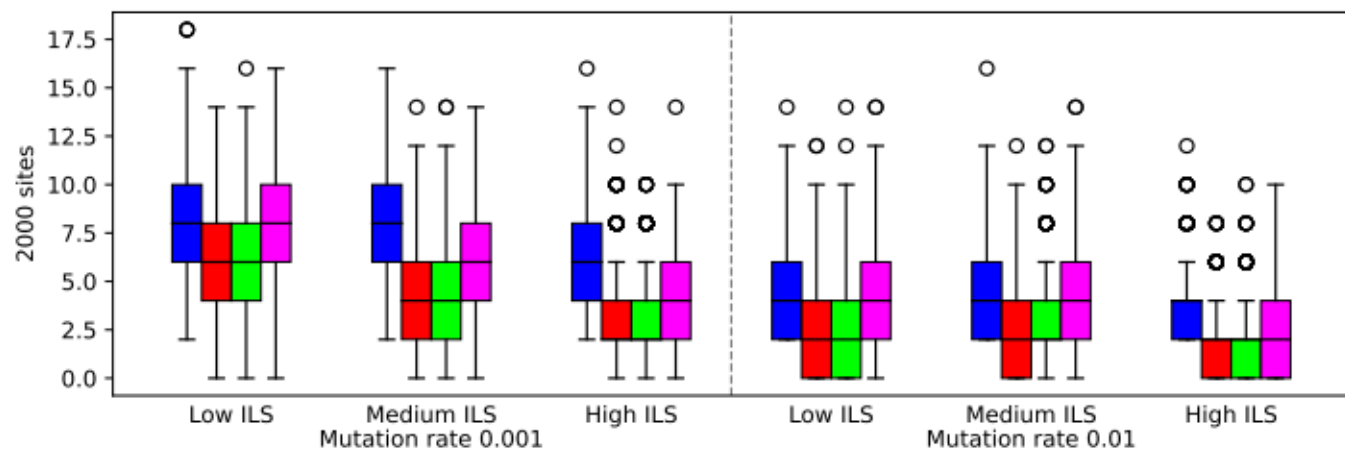
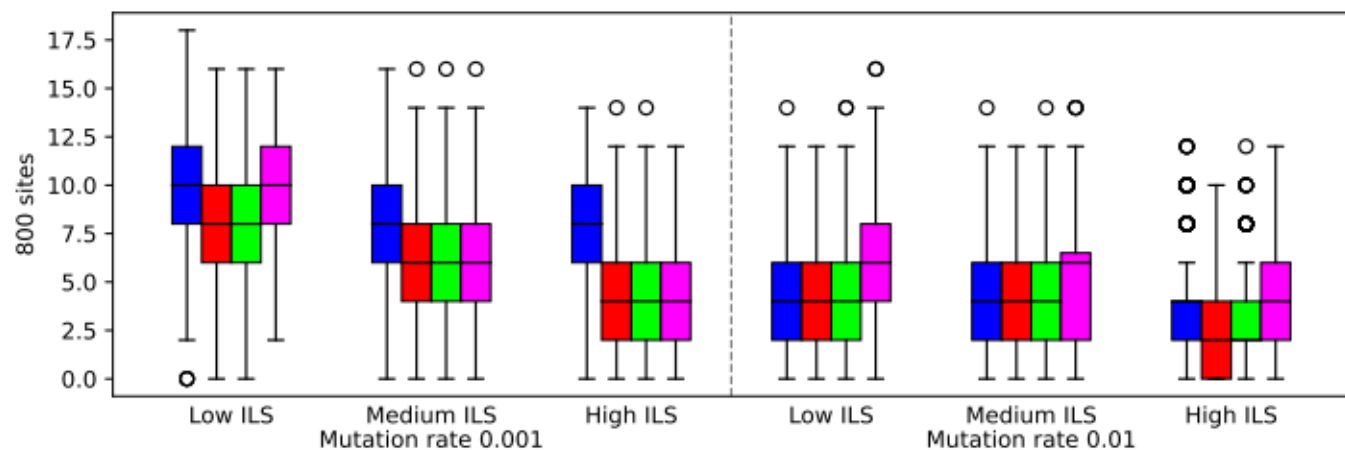
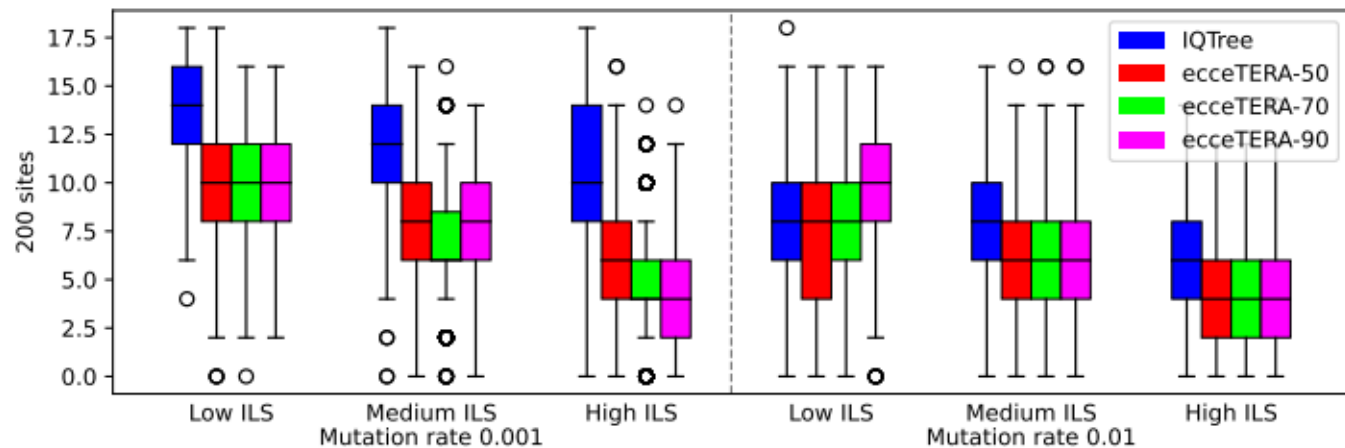
- Collapse low-support branches
- Find refinement of min reconciliation cost

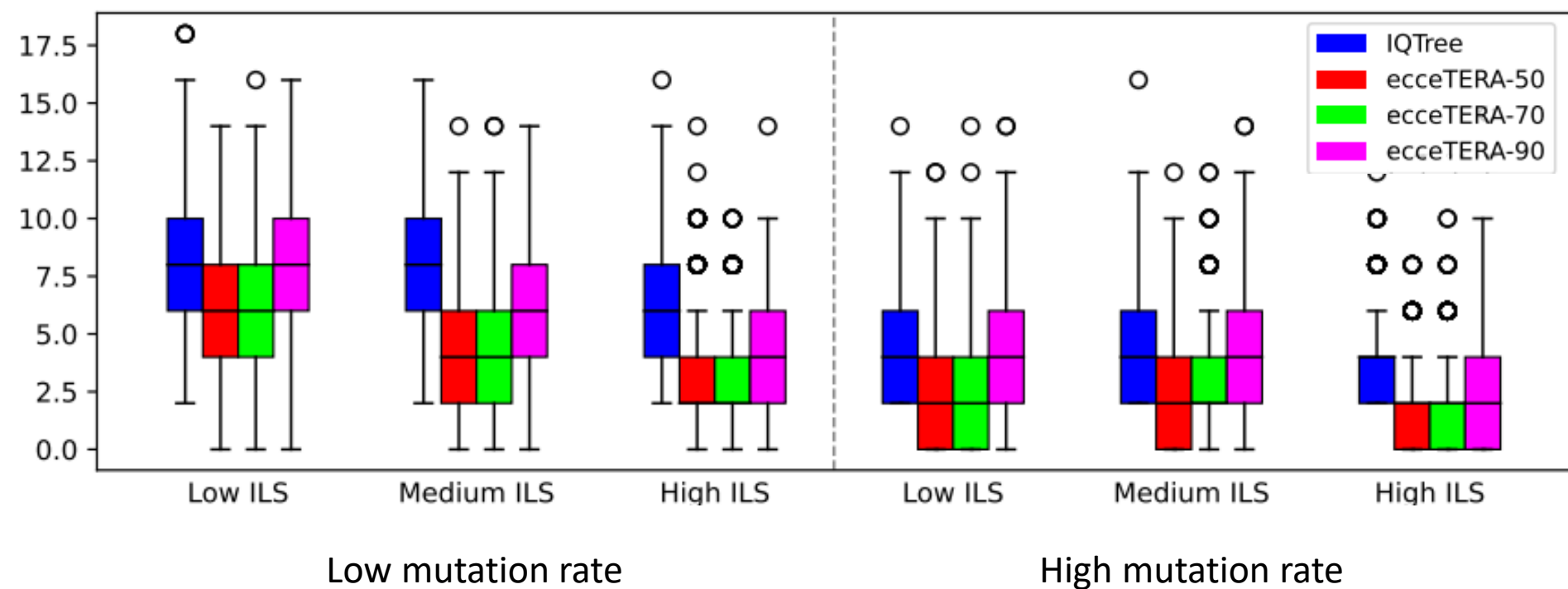
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- Collapse low-support branches (avoid overfitting)
- Find refinement of min reconciliation cost (considers events)

ecceTERA

- Collapse low-support branches (avoid overfitting)
 - Find refinement of min reconciliation cost (considers events)
-
- Tried bootstrap support 50, 70, 90





Conclusions

- Too early to dismiss gene tree correction, even with ILS
- Aggressive branch collapse = bad
- Conservative branch collapse = good
- Analysis only had ILS (all single-copy gene)
- Todo: analyze with dups, losses, transfers, etc.